AI5707 – Pattern Recognition and Machine Learning

Assignment 6: Generation of Data Using 2D Gaussian Mixture Model,
K-Means Clustering and Classification

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1 Introduction

This experiment studies two-dimensional data generation using Gaussian mixture models (GMMs), K-Means clustering, and distortion-based classification. Two classes are generated, each represented by a mixture of three Gaussian components. Isotropic, diagonal, and full covariance matrices are compared. K-Means is fitted separately to the training observations of each class, and held-out observations are classified using minimum squared distortion to the class-specific centroids.

The mixture density is

$$p(\mathbf{x}) = \sum_{i=1}^{K} \pi_i \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i, C_i), \qquad \sum_{i=1}^{K} \pi_i = 1,$$

where $\mathbf{x} = [x_1, x_2]^T$, π_i is the mixture weight, $\boldsymbol{\mu}_i$ is the component mean, and C_i is its covariance matrix.

2 Gaussian Mixture Data Generation

Each class contains 5,000 observations generated from a three-component Gaussian mixture. Class 1 component means are $(-1, -3.5)$, $(0, 0)$, and $(1, 3.5)$. Class 2 is shifted horizontally by four units, giving means $(3, -3.5)$, $(4, 0)$, and $(5, 3.5)$. The mixture weights are $(0.40, 0.35, 0.25)$ for Class 1 and $(0.30, 0.40, 0.30)$ for Class 2.

2.1 Covariance configurations

The three covariance matrices used by the notebook are:

$$C_{\text{iso}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad C_{\text{diag}} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \quad C_{\text{full}} = \begin{bmatrix} 2 & 0.8 \\ 0.8 & 1.5 \end{bmatrix}.$$

Isotropic covariance produces approximately circular clouds. Diagonal covariance allows different variances along the coordinate axes. Full covariance has non-zero off-diagonal entries and can produce tilted elliptical clouds.

2.2 Gaussian sampling

The notebook generates standard-normal values using a polar method based on uniform random values. For full covariance, eigendecomposition $C = V \Lambda V^T$ is used to construct a covariance square-root transform. The transformed samples are shifted by the component mean.

3 Part A: Synthetic Gaussian Data

3.1 Generated distributions

Figure 1 illustrates the generated classes for the isotropic setting. The colours distinguish the three Gaussian components within each class. Histograms for all six components are shown in Figure 2.

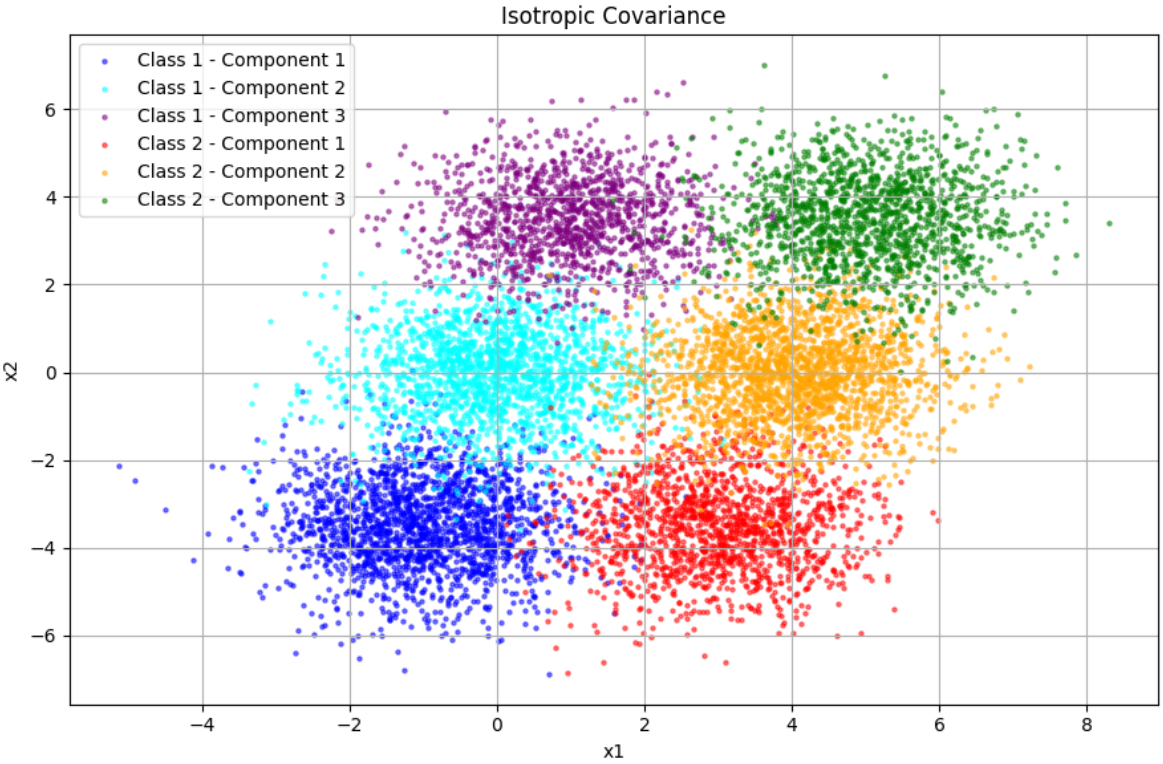


Figure 1: Generated samples for the two classes using isotropic covariance.

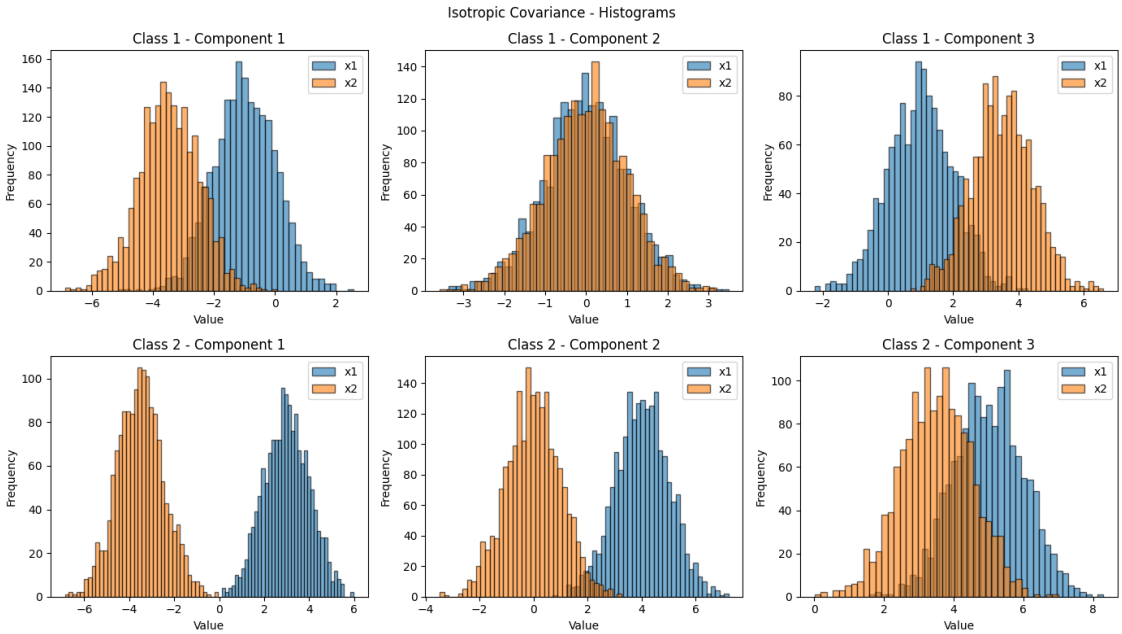


Figure 2: Histograms of the six Gaussian components.

3.2 Train–test split

Each class is split independently in an 80:20 ratio. With 5,000 observations per class, this gives 4,000 training and 1,000 test observations per class. The combined training set contains 8,000 observations and the combined test set contains 2,000 observations.

3.3 K-Means clustering

K-Means is implemented from scratch. It assigns each training point to the nearest centroid using squared Euclidean distance, updates each centroid to the mean of assigned points, and repeats until convergence or the iteration limit is reached. The notebook uses $K = 3$ centroids per class, matching the three Gaussian components used for generation. The plotted regions are within-class nearest-centroid (Voronoi) regions, not the final boundary separating the two classes. Red crosses indicate learned centroids.

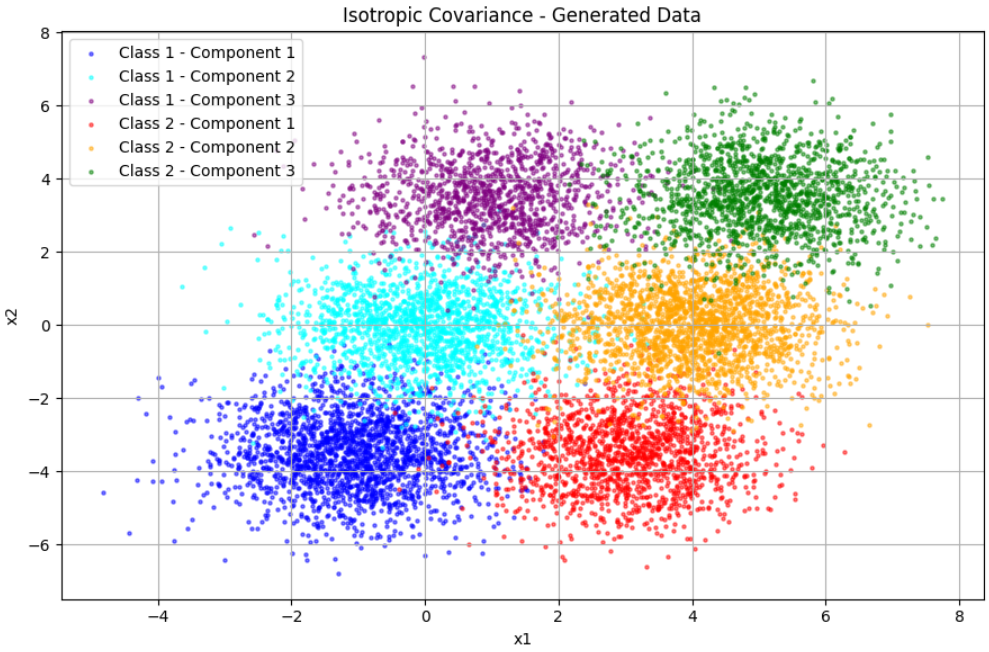


Figure 3: Isotropic case: K-Means regions and centroids for Class 1.

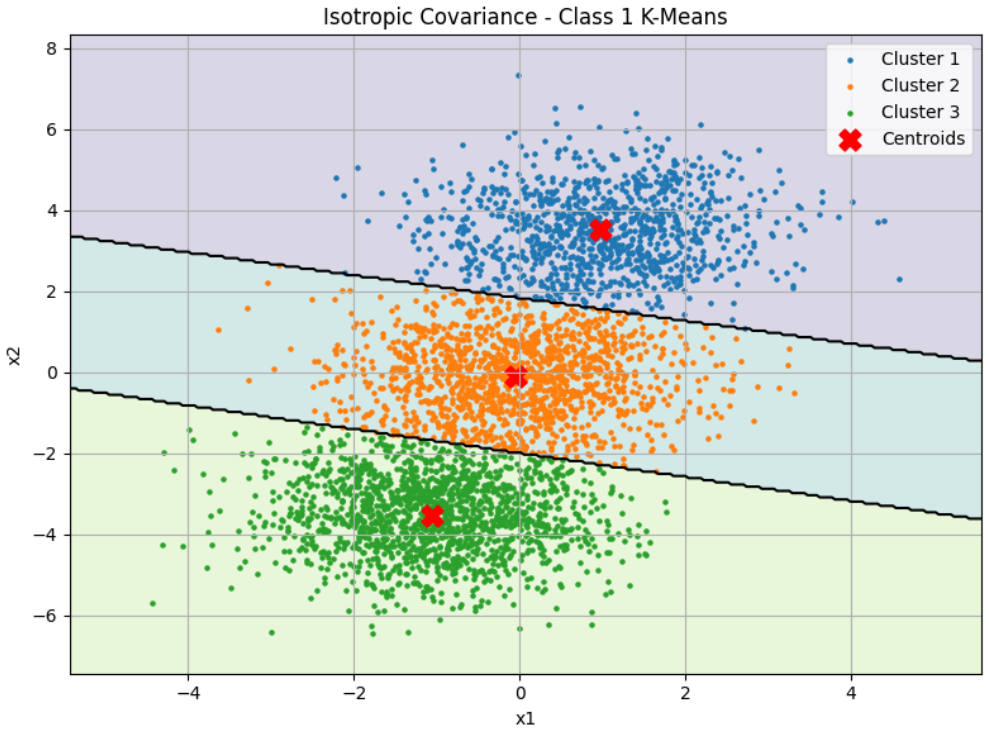


Figure 4: Isotropic case: K-Means regions and centroids for Class 2.

3.4 Distortion-based classification

For a test point $\mathbf{x}$, distortion to class c is the minimum squared distance to its learned centroids:

$$D_c(\mathbf{x}) = \min_k \|\mathbf{x} - \boldsymbol{\mu}_{c,k}\|^2.$$

The predicted class is $\hat{y} = \arg\min_c D_c(\mathbf{x})$. The class decision boundary consists of points satisfying $D_1(\mathbf{x}) = D_2(\mathbf{x})$.

4 Results

4.1 Main test-set performance

The all-cases section of the supplied notebook was executed. Each covariance case has 2,000 test observations in total (1,000 per class). The recorded results are summarized in Table 1.

Table 1: Test-set performance for all covariance configurations

Covariance type	Correct / total	Accuracy	Error rate
Isotropic	1,959 / 2,000	97.95%	2.05%
Diagonal	1,675 / 2,000	83.75%	16.25%
Full	1,858 / 2,000	92.90%	7.10%

4.2 Confusion matrices

Rows indicate the true class and columns indicate the predicted class. The matrices below are from the recorded notebook execution.

Table 2: Isotropic covariance confusion matrix

	Predicted 1	Predicted 2
True Class 1	978	22
True Class 2	19	981

Table 3: Diagonal covariance confusion matrix

	Predicted 1	Predicted 2
True Class 1	847	153
True Class 2	172	828

Table 4: Full covariance confusion matrix

	Predicted 1	Predicted 2
True Class 1	935	65
True Class 2	77	923

4.3 Test predictions and cluster regions for all cases

The following figures show the K-Means regions for each class and the distortion-based classification of the combined test set. In each prediction plot, the decision boundary is based on the minimum squared distances to the centroids of both class models.

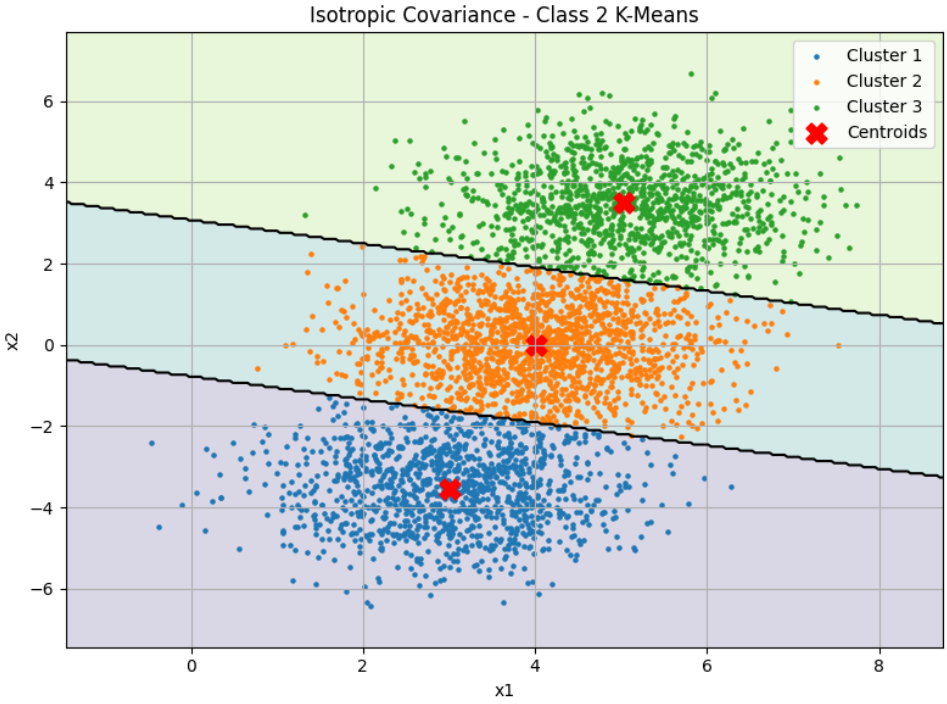


Figure 5: Distortion-based test classification for isotropic covariance.

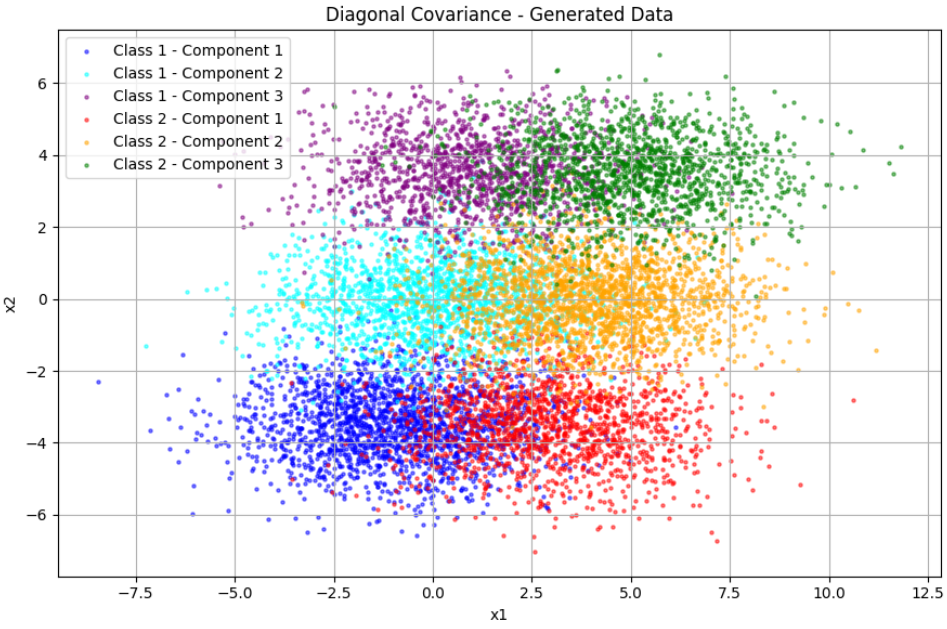


Figure 6: Diagonal case: K-Means regions and centroids for Class 1.

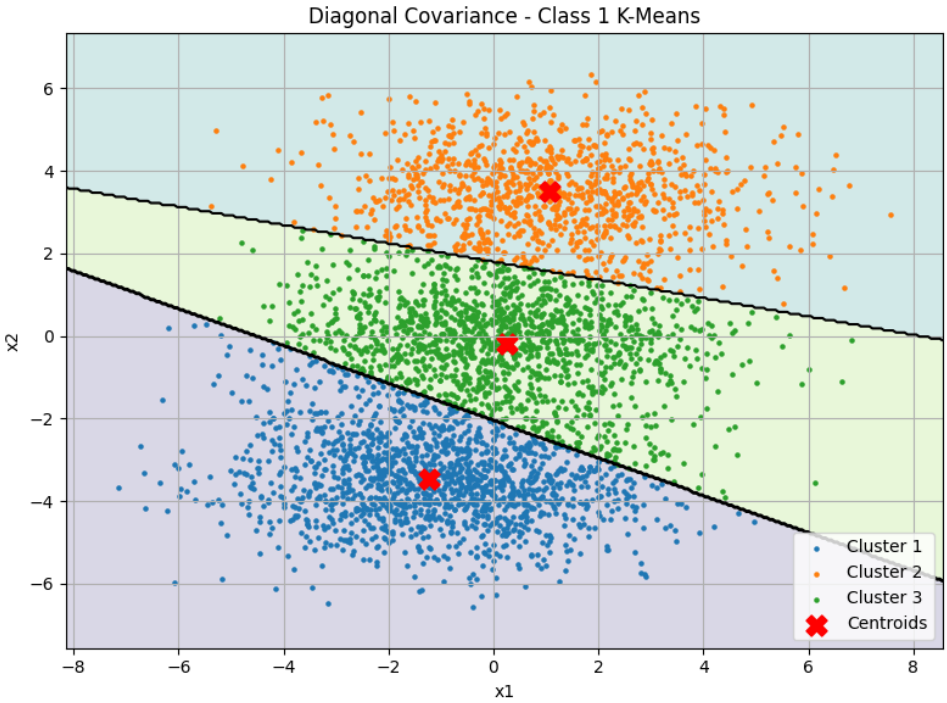


Figure 7: Diagonal case: K-Means regions and centroids for Class 2.

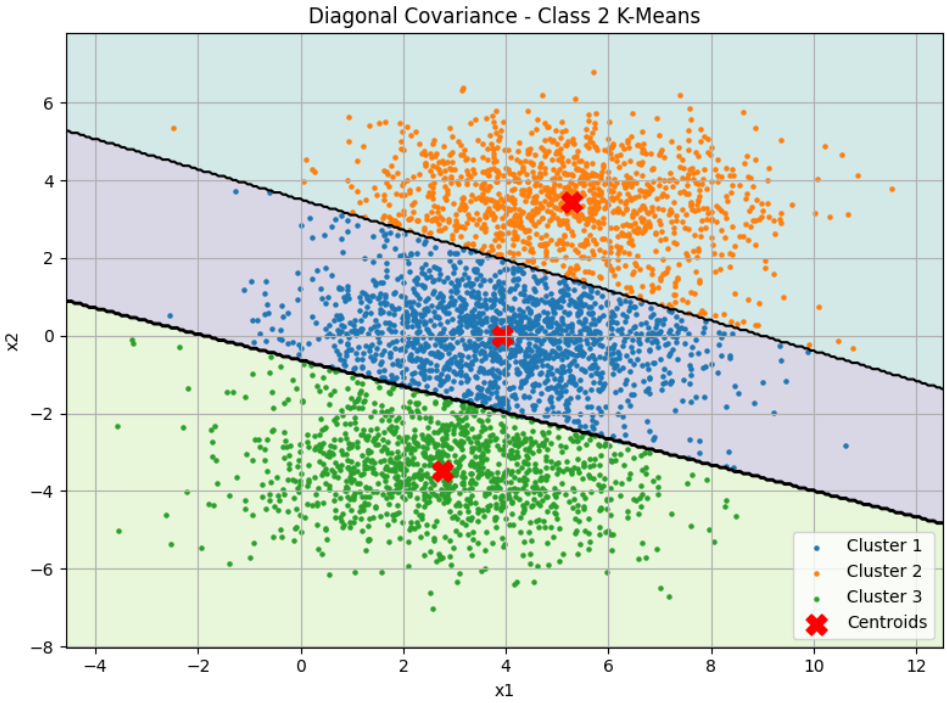


Figure 8: Distortion-based test classification for diagonal covariance.

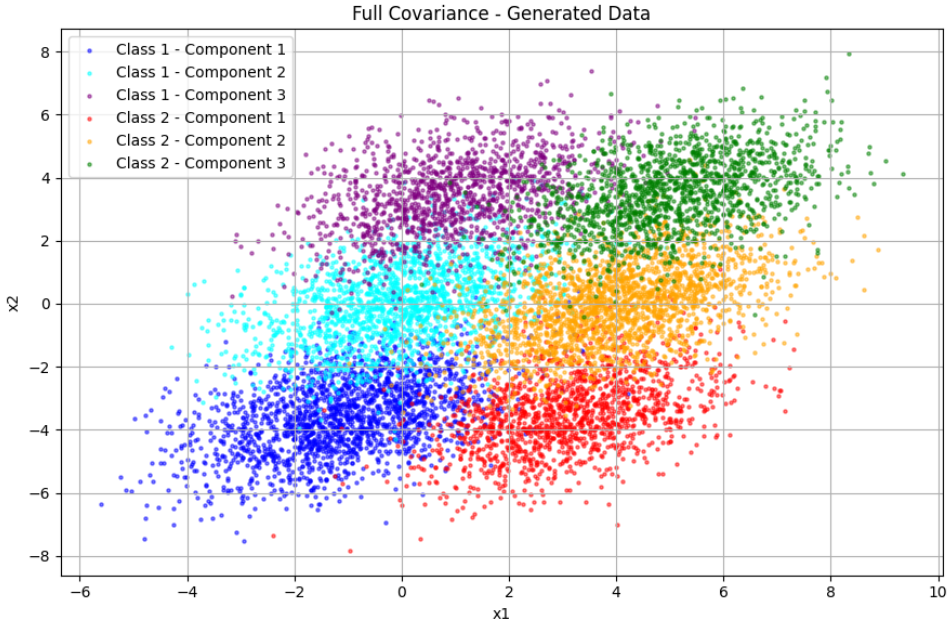


Figure 9: Full covariance case: K-Means regions and centroids for Class 1.

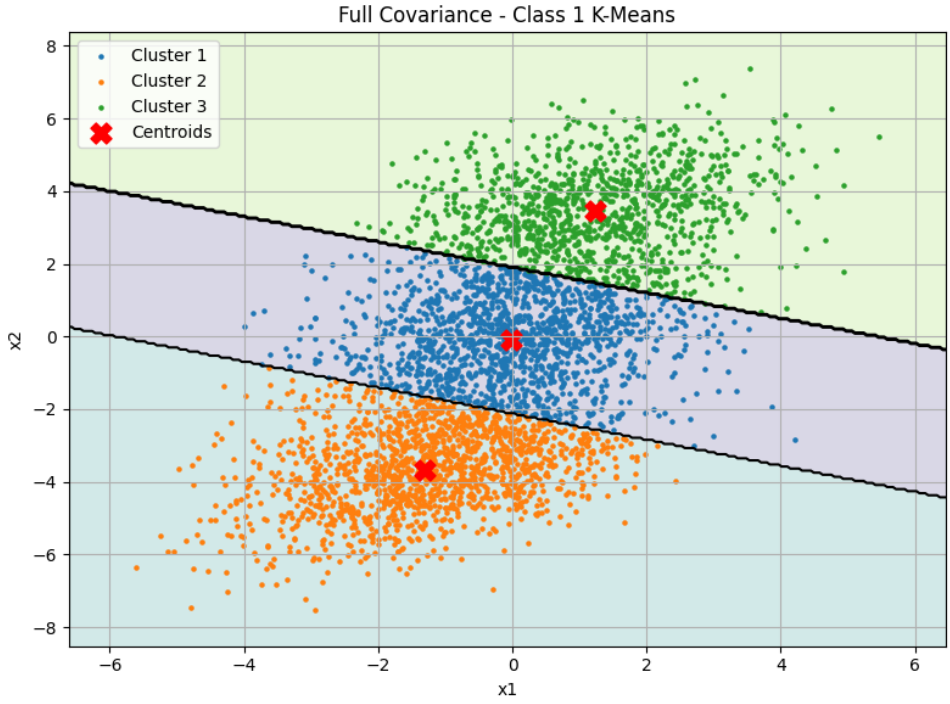


Figure 10: Full covariance case: K-Means regions and centroids for Class 2.

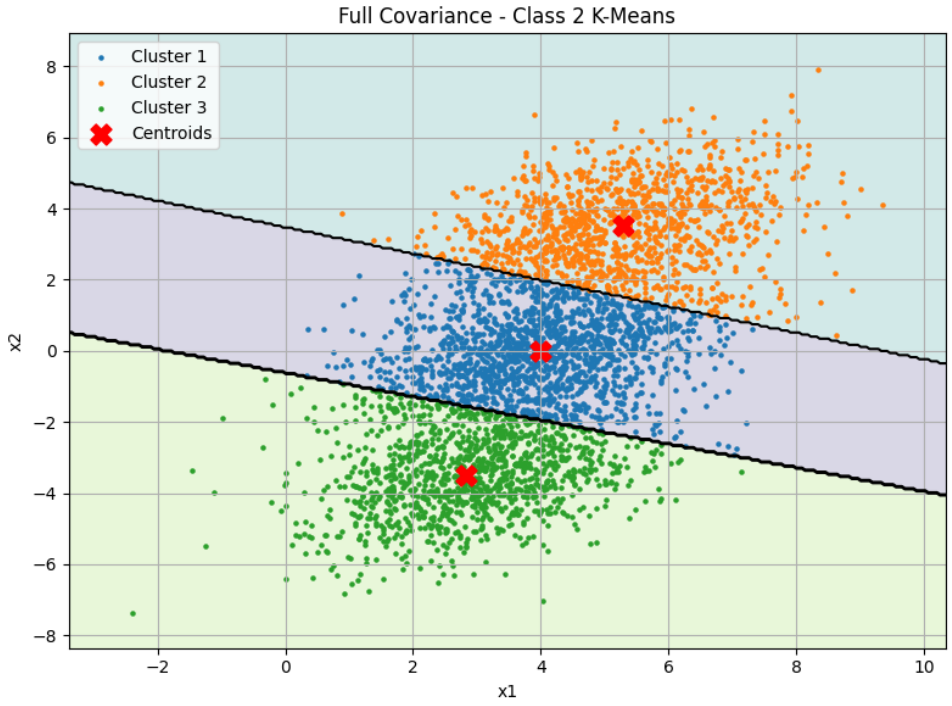


Figure 11: Distortion-based test classification for full covariance.

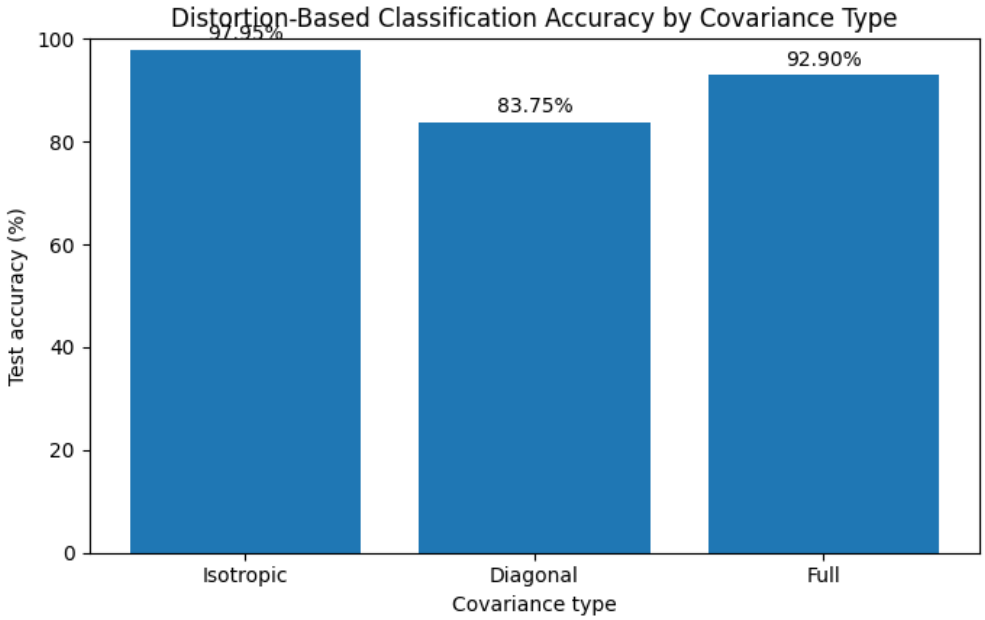


Figure 12: Test accuracy comparison across the three covariance configurations.

5 Answers to Assignment Questions

Question 1: Choose K, π_i, μ_i, and C_i empirically.
Three Gaussian components are used per class. The means and mixture weights are described in Section 2, and the three covariance matrices are specified in Section 2.1.

Question 2: Experiment with isotropic, diagonal, and full covariance matrices.
All three covariance configurations are executed in the final experiment section of the notebook and compared in Table 1.

Question 3: Plot points belonging to different clusters using different colours.
The generated-data plot uses different colours to distinguish the three Gaussian components within each class.

Question 4: Generate data for two classes, each represented by a mixture density.
Two classes of 5,000 observations each are generated, with three Gaussian components per class and horizontal separation.

Question 5: Divide data 80:20, perform K-Means on 80%, and plot cluster boundaries.
Each class has 4,000 training and 1,000 test observations. K-Means uses three centroids per class. The cluster-region figures show the learned within-class partitions.

Question 6: Classify the remaining 20% using distortion.
Each test sample is assigned to the class with the smaller minimum squared centroid distance. The recorded accuracies are 97.95%, 83.75%, and 92.90% for isotropic, diagonal, and full covariance respectively.

Question 7: Report observations.
Covariance affects cluster geometry and overlap. K-Means learns class-specific prototypes, and distortion classification assigns each test point to the class whose nearest centroid is closer.

6 Observations

1. The Gaussian-mixture generator produces three components for each of the two classes.
2. Isotropic covariance produces roughly circular clouds; diagonal covariance allows unequal axis-wise variance; full covariance allows correlated dimensions.
3. The dataset contains 5,000 samples per class and uses an 80:20 train–test split.
4. K-Means uses three centroids per class, while the classification step compares distances across both class-specific centroid sets.
5. Isotropic covariance achieved the highest accuracy in this run (97.95%); diagonal covariance achieved 83.75%; full covariance achieved 92.90%.
6. The accuracy differences reflect the distribution geometry and the minimum-centroid-distance decision rule.
7. Results may vary with random train–test splits and centroid initialization.

7 Conclusion

Two two-dimensional Gaussian-mixture classes were generated and evaluated under isotropic, diagonal, and full covariance assumptions. The data were split in an 80:20 ratio, K-Means was fitted independently to each class, and held-out observations were classified using minimum squared distortion. In the recorded run, isotropic covariance achieved 97.95% accuracy, diagonal covariance 83.75%, and full covariance 92.90%. The experiment demonstrates that covariance structure influences cluster geometry and the performance of a centroid-based classifier.